Suggested problems: 1-10

Problem 1: Suppose X and Y are jointly discrete random variables with joint probability function $f(x, y)$ given by

$f(x,y)$	$y=1$	$y=2$	$y=3$	$y=4$
$x=1$	0.1	0	0.1	0
$x=2$	0.3	0	0.1	0.2
$x=3$	0	0.2	0	0

Compute the marginal probability functions of X and Y.

Problem 2: Suppose X and Y are jointly continuous random variables with joint probability density function $f(x, y)$ given by

$$f(x,y) = \begin{cases} \frac{21}{4}x^2 y & : x^2 \leq y \leq 1, \\ 0 & : \text{else.} \end{cases}$$

Compute the marginal probability density functions of X and Y.